ConsensusMetaDA: R package for microbiome analysis using multiple algorithm

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Introduction to ConsensusMetaDA

ConsensusMetaDA is an R package for microbiome analysis using multiple algorithms - reaching consensus. No one tool perform best DA anlayes on metaganome data. Hence it needs a consensus approach to obtain more robust differential abundant microbiome. ConsensusMetaDA helps achieve this using five popularly used tools. ConsensusMetaDA uses popular biom format and samples table as input to perform DA along with it generates standard microbiome visualization plots such as Alpha Diversity, Beta Diversity, Rarefaction curve, Bidirectional plot and Scale plot at various taxa levels.

```r
# install supporting packages

# ADAPT R package requires R version 4.4 hence need all R updated to R v4.4.2

# One line per package
if (!require("ggplot2", quietly = TRUE)) install.packages("ggplot2")
```

```r
if (!require("BiocManager", quietly = TRUE)) install.packages("BiocManager")
if (!require("phyloseq", quietly = TRUE)) BiocManager::install("phyloseq")
if (!require("DESeq2", quietly = TRUE)) BiocManager::install("DESeq2")
if (!require("ALDEx2", quietly = TRUE)) BiocManager::install("ALDEx2")
if (!require("edgeR", quietly = TRUE)) BiocManager::install("edgeR")
if (!require("ADAPT", quietly = TRUE)) BiocManager::install("ADAPT")
if (!require("metagenomeSeq", quietly = TRUE)) BiocManager::install("metagenomeSeq")

if (!require("ConsensusMetaDA", quietly = TRUE)) {
  if (!require("remotes", quietly = TRUE)) install.packages("remotes")
  remotes::install_github("kmanoharan01/ConsensusMetaDA")
}
```

Load Libraries

```r
# Load the packages
library(ggplot2)
library(phyloseq)
library(DESeq2)
library(ALDEx2)
library(edgeR)
library(metagenomeSeq)
library(ADAPT)


# load ConsensusMetaDA

library(ConsensusMetaDA)
```

ConsensusMetaDA examples

Begin by first installing and loading the `consensusDE` library. To illustrate functionality of `consensusDE`, we will utilise RNA-seq data from the `airway` and annotation libraries as follows. Begin by installing and attaching data from these libraries as follows:

```r
#build_OTU_counts(GWMC_HOT_COLD)


# OTUs_multi_DE(GWMC_HOT_COLD)

#OTU_plots(GWMC_HOT_COLD)
```

Building a phyloseq format

A Generates summarized experiment utilizing inputs such as biom file, sample list, and taxa information and creates a buildOTU object. To build a summarized experiment object ConsensusMetaDA requires a biom file and sample grouping table with location path. Taxa level by accepts 7 taxa level. This function is incorporated utilizing the phyloseq (v1.50.0) package (McMurdie & Holmes, 2013). It also can take filtering parameters such as abundance_threshold, prevalence_threshold, rarity_threshold and variance_threshold

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

filtering rarity format

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

filtering abundance format

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

filtering prevalence format

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

filtering variance format

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

Performing differential abundance

Performs DA analysis on all possible pairs of "groups" on the provided sample table using all available methods (currently edgeR, DESeq2, metagenomeSeq, ADAPT and ALDEx2). Also, OTUs_multi_DE produces a Venn diagram and upset plots for all comparisons. We have also implemented rank-sum (Guo, Zhao, Ye, Sheng, & Shyr, 2014) to perform p-value-based ranking in the summary table. The function creates a table for each comparison. The summary table contains statics including "p_union", "p_intersect" and "rank_sum". The summary table provides results from three tools in one format hence easy to compare. "AveExpre" is the mean value of normalized average counts.

Running `OTUs_multi_DE()` will perform DE analysis on all possible pairs of "groups" and save these results as a simple list of "merged" - being the merged results of "deseq", "voom" and "edger" into one table, as well as the latter three as objects independently. The data frame is sorted by the `rank_sum`. The following columns are included:

- `ID` - Identifier
- `AveExpr` - Average Expression (average of edgeR, DESeq2 and voom)
- `LogFC` - Log2 Fold-Change, also known as a log-ratio (average of edgeR, DESeq2 and voom)
- `LogFC_sd` - Log2 Fold-Change standard deviation of LogFC (average)
- `edger_adj_p` - EdgeR p-value adjusted for multiple hypotheses
- `deseq_adj_p` - DESeq2 p-value adjusted for multiple hypotheses
- `voom_adj_p` - Limma/voom p-value adjusted for multiple hypotheses
- `edger_rank` - rank of the p-value obtained by EdgeR
- `deseq_rank` - rank of the p-value obtained by DESeq2
- `voom_rank` - rank of the p-value obtained by Limma/voom
- `rank_sum` - sum of the ranks from edger_rank, voom_rank, rank_sum
- `p_intersect` - the largest p-value observed from all methods tested.
  - This represents the intersect when a threshold is set on the p_intersect column
- `p_union` - the smallest p-value observed from all methods tested.
  - This represents the union when a threshold is set on the p_union column To access the merged results:

UpSet Plot from Differential abundance

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

Venn Diagram from Differential abundance

```
GWMC_HOT_COLD
```

```
## phyloseq-class experiment-level object
## otu_table()   OTU Table:         [ 4401 taxa and 798 samples ]
## sample_data() Sample Data:       [ 798 samples by 2 sample variables ]
```

Plotting functions

When performing DA analysis, a series of plots (currently 5) can be generated and saved as .pdf files in a plot directory provided to `multi_de_pairs()` with the parameter: `plot_dir = "/path/to/save/pdfs/`. See `?multi_de_pairs` for description.

In addition, each of the 5 plots can be plotted individually using the `OTU_plots` function. See `?OTU_plots` for description, which provides wrappers for 5 different plots. Next we will plot each of these using the example data.

rarefaction curve

Plot the number of reads that mapped to the transcriptome of each sample. The sample numbers on the x-axis correspond to the sample row number in the summarizedExperiment built, accessible using `colData(airway)`. Samples are coloured by their "group".

```
#OTU_plots(build_OTU_counts_output = GWMC_HOT_COLD
 #          )
```

Alpha Diversity

```
#OTU_plots(build_OTU_counts_output = GWMC_HOT_COLD
 #          )
```

Beta Diversity/Principle Coordinates Analysis

```
#OTU_plots(build_OTU_counts_output = GWMC_HOT_COLD
 #          )
```

Bidirectional Plot

Residuals for the RUV model can be plotted as follows:

```
OTU_plots(build_OTU_counts_output = GWMC_HOT_COLD
           )
```

Citing results that use `ConsesnsusMetaDA`

When using this package, please cite ConsesnsusMetaDA as follows and all methods used in your analysis.

For consensus DE:

```
#citation("ConsesnsusMetaDA")
```

- When using DESeq2 (also check package reference suggestions)
  - Love, M.I., Huber, W., Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2 Genome Biology 15(12):550 (2014)
- When using edgeR (also check package reference suggestions)
  - Robinson MD, McCarthy DJ and Smyth GK (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. Bioinformatics 26, 139-140
  - McCarthy DJ, Chen Y and Smyth GK (2012). Differential expression analysis of multifactor RNA-Seq experiments with respect to biological variation. Nucleic Acids Research 40, 4288-4297
- When using ALDEx2 (also check package reference suggestions)
  - Fernandes, A. D., Macklaim, J. M., Linn, T. G., Reid, G., & Gloor, G. B. (2013). ANOVA-Like Differential Expression (ALDEx) Analysis for Mixed Population RNA-Seq. PLOS ONE, 8(7), e67019. doi:10.1371/journal.pone.0067019
- When using ADAPT (also check package reference suggestions)
  - Wang, M., Fontaine, S., Jiang, H., & Li, G. (2024). ADAPT: Analysis of Microbiome Differential Abundance by Pooling Tobit Models. Bioinformatics, 40(11). doi:10.1093/bioinformatics/btae661
- When using MetagenomeSeq (also check package reference suggestions)

– Paulson JN, O. N., Braccia DJ, Wagner J, Talukder H, Pop M, Bravo HC (2013). metagenomeSeq: Statistical analysis for sparse high-throughput sequncing. Bioconductor package. Retrieved from http://www.cbcb.umd.edu/software/metagenomeSeq.